

mic — Project Reference

A quiet companion for daily reading: the app, the content pipeline, the landing page, and the feedback API.

Last updated: August 2026 · App version: 1.0.3+4 · Flutter 3.44.8 / Dart ^3.12.2

Repos: `sir-mic/pilgrim` (app) · `sir-mic/path2mic` (landing page)

Live site: <https://sir-mic.github.io/path2mic/>

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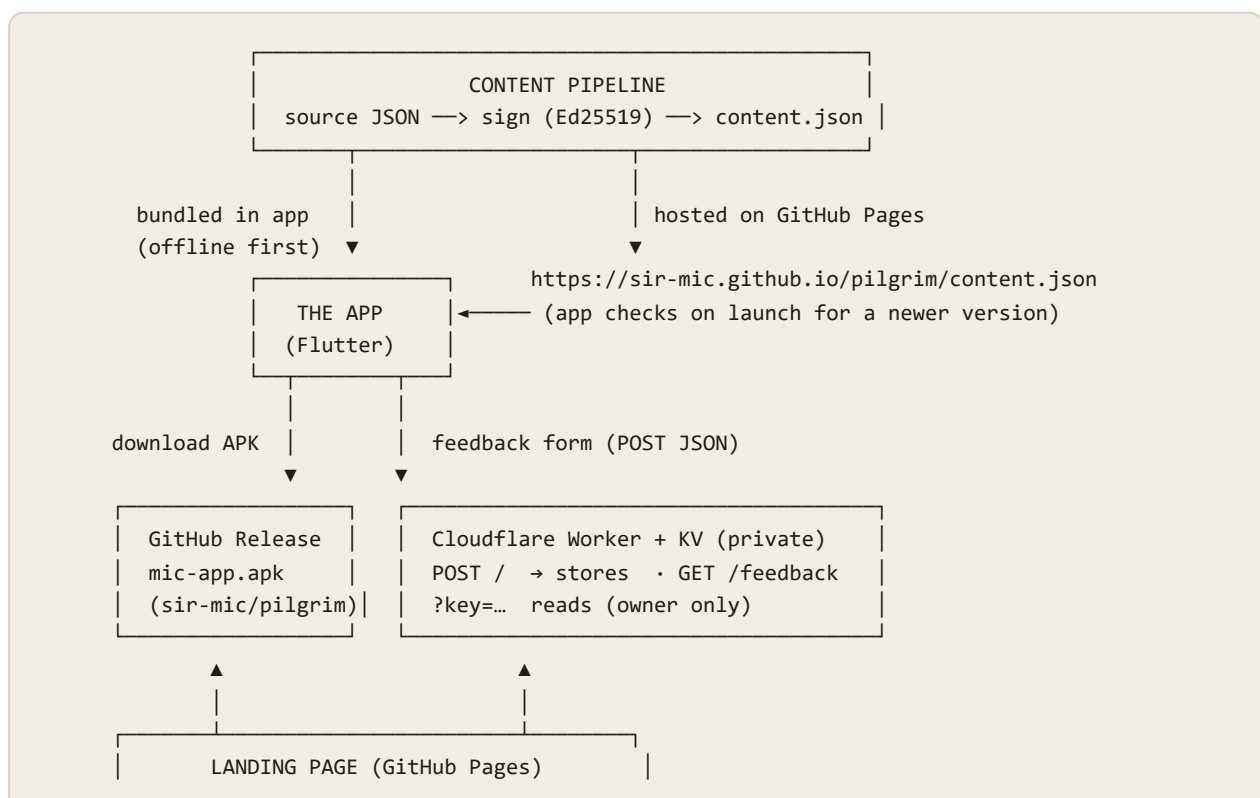
1. Overview & system diagram

mic (internally *Pilgrim*, package `pilgrim_app`) is a free Android app for reading the Bible a little at a time. It ships reading plans (one chapter a day), a daily reflection prompt, a journal, progress tracking, and an optional quiet daily reminder.

The whole project is split across four pieces:

- **App** — Flutter/Android app that stores everything locally in SQLite.
- **Content** — a signed, data-driven "content bundle" (plans, prompts, notification copy) that the app can refresh without an app update.
- **Landing page** — a GitHub Pages site with the download button and the feedback form.
- **Feedback API** — a Cloudflare Worker + KV that stores feedback submissions privately.

System diagram



`https://sir-mic.github.io/path2mic/`

Everything the user creates (journal, progress, settings) lives only on the device, in the app's private SQLite database. There is no server-side account and no backup/export feature yet.

2. Folder / repository map

Folder	Git repo	What it is
<code>app/</code>	<code>sir-mic/pilgrim</code>	The Flutter Android app (<code>pilgrim_app</code>).
<code>content/</code>	<i>not in git</i>	Content source, signing tool and private key . Local only.
<code>admin/</code>	<i>not in git</i>	Flutter web admin tool for editing and publishing content.
<code>site/</code>	<code>sir-mic/path2mic</code>	Landing page + Cloudflare Worker code.

Security: `content/keys/private_key.pem` is the Ed25519 signing key for content bundles. It is *not* in any git repository and must never be committed. Only the public key ships in the app.

3. The Android app

3.1 Stack

Piece	Choice	Role
Framework	Flutter 3.44.8 (stable)	UI + Android target
Language	Dart ^3.12.2	App language
State	flutter_riverpod ^2.6.1	Providers, reactive state
Local DB	drift ^2.34.3 + drift_flutter	SQLite (all user data)
Notifications	flutter_local_notifications ^22.2.0, timezone, flutter_timezone	Daily quiet reminder (exact alarms)
Content security	cryptography ^2.9.0	Ed25519 signature verification
Networking	http ^1.6.0	Remote content refresh
Deep links	url_launcher ^6.3.2	"Send feedback" → opens landing page
Dates	intl ^0.20.3	Date formatting
Content	pilgrim_content (path <code>../content</code>)	Models, validation, verification
Font	Lora (variable)	App + landing page typeface

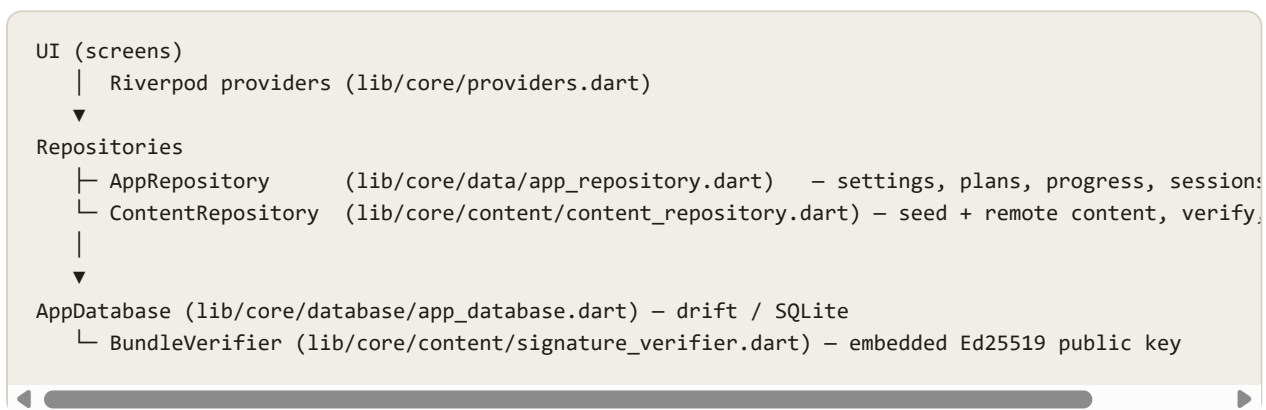
Build	<code>build-apk.ps1</code>	Builds and renames the release APK
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Android identity: `applicationId com.pilgrim.pilgrim_app`, debug-signed release builds, `version 1.0.3+4` (name 1.0.3, code 4). App display name: **mic**.

3.2 Startup flow (`lib/main.dart`)

1. `main()` initialises the notification service, then runs the app inside a Riverpod `ProviderScope`.
2. The bootstrap provider seeds the bundled content (verifies the signature, reconciles into the DB) and restores the saved theme.
3. If not onboarded → `OnboardingScreen`; otherwise → `HomeShell`.
4. After first frame, `HomeShell` silently checks for a remote content update and re-schedules the daily reminder if enabled.

3.3 Architecture



3.4 Database schema (drift, `schemaVersion 2`, file `pilgrim.db`)

Table	Holds
<code>content_meta</code>	Which content-bundle version is loaded (<code>id</code> , <code>version</code> , <code>updatedAt</code>).
<code>plans</code>	Reading plans (<code>slug</code> , title, description, kind, total days).
<code>plan_days</code>	One row per plan day; <code>readings</code> is a JSON array.
<code>reflection_prompts</code>	Daily reflection prompt pool.
<code>notification_messages</code>	Calm reminder message pool.
<code>sessions</code>	Journal entries — store an immutable snapshot of the day's readings so later content changes never rewrite history.
<code>settings_table</code>	Key/value settings (see keys below).
<code>reading_progress</code>	Partial progress through today's readings; keyed by (<code>planSlug</code> , <code>date</code>) so it expires each day.

Settings keys: `planSlug`, `onboarded`, `reminderEnabled`, `reminderHour`, `reminderMinute`, `themeMode`, `manifestUrl`, and `progress_<planSlug>` (days completed this run).

3.5 Screens

- **Onboarding** — first-run flow; picks a plan.
- **Home / Today** — today's reading, prompt, completion.
- **Reading flow** — one chapter at a time with rest/resume progress.
- **Journal** — timeline of completed sessions, searchable, with detail view.
- **Progress** — stats: completed days, sessions per plan, completed plans.
- **Settings** — daily reminder (+ time, exact alarms, Autostart hint), theme, current plan / restart plan, About with *Send feedback*.

3.6 Reminder

- Scheduled daily via `flutter_local_notifications` with exact alarms (Android 12+).
- Time zone handled with `timezone` + `flutter_timezone`.
- Re-scheduled on every launch if enabled; message picked from the content bundle.

3.7 Send feedback (Settings → About)

A `ListTile` in Settings opens `https://sir-mic.github.io/path2mic/#feedback` in the external browser via `url_launcher` — where the user can submit a private feedback message.

4. Content pipeline & signing

Reading content is *data, not code*. A signed JSON bundle carries plans, prompts and notification messages. The app ships a copy (offline first) and can pull a newer bundle from the web — no app update required.

4.1 The `pilgrim_content` package (`content/`, local path dependency)

File	Purpose
<code>lib/src/models.dart</code>	Content models (<code>SignedBundle</code> , plans, readings...).
<code>lib/src/validation.dart</code>	Content validation.
<code>lib/src/canonical_json.dart</code>	Canonical JSON so signatures are stable.
<code>lib/src/signing.dart</code>	Ed25519 sign/verify helpers.
<code>lib/src/book_table.dart</code> , <code>built_plans.dart</code> , <code>mcheyne_parser.dart</code> , <code>default_copy.dart</code>	Plan/verse helpers and defaults.
<code>bin/pilgrim_build.dart</code>	CLI: <code>keygen</code> and <code>build</code> .

4.2 Build commands

```
cd content
dart run pilgrim_build keygen      # new Ed25519 key pair → keys/private_key.pem (+ printed public key)
```

```
dart run pilgrim_build build # validate + sign source/mcheyne-plan.json
# → build/content.json and ../app/assets/content/content.json
```

4.3 Verification & reconcile (in the app)

- The app embeds the **public** key (`signature_verifier.dart`).
- On first run it verifies the bundled asset and seeds the DB.
- On launch it fetches `https://sir-mic.github.io/pilgrim/content.json`, verifies the signature, and reconciles if the version is newer. Failures are silent — the app keeps its last-known-good local copy and stays usable offline.
- Reconcile upserts plans, replaces prompts/messages; user progress and journal are never touched.

5. Admin tool

A small Flutter *web* app (`admin/`, local only) for maintaining content. It has four parts:

File	Purpose
<code>lib/main.dart</code>	App shell.
<code>lib/src/plans_tab.dart</code>	Edit reading plans/days.
<code>lib/src/copy_tab.dart</code>	Edit prompts and notification copy.
<code>lib/src/publish_tab.dart</code>	Trigger a build + publish of <code>content.json</code> .
<code>lib/src/github_publish.dart</code>	Publishes the bundle to the <code>sir-mic/pilgrim</code> GitHub Pages manifest URL.

Publishing updates `https://sir-mic.github.io/pilgrim/content.json`, which installed apps pick up automatically.

6. Landing page

`site/index.html` is a single self-contained page (CSS + JS inline) hosted on GitHub Pages at `https://sir-mic.github.io/path2mic/`. Pushing to `main` auto-publishes.

6.1 Theme

Token	Value	Use
<code>--bg</code>	<code>#161310</code>	Page background
<code>--surface</code>	<code>#201B17</code>	Cards / forms
<code>--border</code>	<code>#342F29</code>	Borders
<code>--gold</code>	<code>#C9A876</code>	Accent
<code>--cream</code>	<code>#EDE6D8</code>	Primary text

--muted / --dim	#A89E8C / #716C64	Secondary / tertiary text
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Typeface: Lora (Google Fonts). Sections: nav, hero (phone mockup), about, download, feedback.

6.2 Download button

Points to `https://github.com/sir-mic/pilgrim/releases/latest/download/mic-app.apk` — always the newest published release asset.

6.3 Feedback form

A mobile-friendly form (name, type bug/suggestion, message) that POSTs JSON to the Cloudflare Worker URL wired into `const API_URL`. Feedback is **never shown publicly** on the page.

7. Feedback API (Cloudflare Worker + KV)

Live worker: `https://divine-glitter-11cefeedback.mikeharoldhandig12.workers.dev` · Source: `site/worker/feedback-worker.js` · Deploy steps: `site/worker/README.md`.

7.1 Endpoints

Method + path	Behaviour
POST /	Stores <code>{ name, type ("bug" "suggestion"), message, date }</code> in KV. Returns <code>201</code> .
GET /feedback?key=...	Returns <code>{ "items": [...] }</code> newest-first. 403 without the secret key .
OPTIONS	<code>204</code> with CORS headers (writes are open so the site can submit).

7.2 Setup summary

- Cloudflare Worker with a KV namespace binding named `FEEDBACK`.
- Text variable `FEEDBACK_SECRET` = your private read key.
- Read feedback at `/feedback?key=YOUR_SECRET` (browser or `curl`).

```
# submit
curl -X POST https://...workers.dev -H "Content-Type: application/json" \
  -d '{"name":"Tester","type":"bug","message":"Reminder fired early"}'

# read (private)
curl "https://...workers.dev/feedback?key=YOUR_SECRET"
```

8. Build & release process

8.1 Building the APK

```
cd app
powershell -ExecutionPolicy Bypass -File .\build-apk.ps1
```

```
# → build\app\outputs\flutter-apk\mic-app.apk (=58 MB)
```

The script runs `flutter build apk --release` then renames the artifact to `mic-app.apk` (Flutter always emits `app-release.apk` internally).

8.2 Signing & updates (data safety)

- Release builds are signed with the **debug keystore** (`~/.android/debug.keystore`).
- Updating keeps all data (journal, progress, settings) *as long as* the new APK has the same signature and a version code $\geq$ installed.
- Building on a *different machine* produces a different debug keystore → signature mismatch → install fails and data would be lost after a forced uninstall. Build on this machine.
- Version lives in `app/pubspec.yaml` (`version: name+build`). Bump it on every release.

8.3 Publishing a release

1. Build `mic-app.apk` (above).
2. On `sir-mic/pilgrim` → Releases → create/edit release with tag matching the app version.
3. Attach `mic-app.apk` (the landing page download link expects that filename).

9. URL quick-reference

Thing	URL
App repo	<code>https://github.com/sir-mic/pilgrim</code>
Landing page repo	<code>https://github.com/sir-mic/path2mic</code>
Landing page (live)	<code>https://sir-mic.github.io/path2mic/</code>
Feedback anchor	<code>https://sir-mic.github.io/path2mic/#feedback</code>
APK download	<code>https://github.com/sir-mic/pilgrim/releases/latest/download/mic-app.apk</code>
Content manifest	<code>https://sir-mic.github.io/pilgrim/content.json</code>
Feedback Worker	<code>https://divine-glitter-11cefeedback.mikeharoldhandig12.workers.dev</code>
Feedback read (private)	<code>.../feedback?key=YOUR_SECRET</code> (403 without key)

10. FAQ & troubleshooting

Will updating the app lose my journal/notes?

No — if the new APK has the same signature and an equal-or-higher version code, Android upgrades in place and keeps all data. Data is only lost on uninstall, factory reset, or a signature mismatch (see §8.2).

The download button on the site returns nothing.

It only works once a release exists on `sir-mic/pilgrim` with an asset named `mic-app.apk`.

How do I see feedback submissions?

Visit `https://divine-glitter-11cefeedback.mikeharoldhandig12.workers.dev/feedback?key=YOUR_SECRET` in a browser, or download as JSON with `curl`.

Feedback form says the worker URL isn't set.

In `site/index.html`, the `API_URL` constant must be the live worker URL (it currently is). A `YOUR_WORKER_URL_HERE` placeholder means it wasn't wired up.

I changed content — how do installed apps get it?

Rebuild with `pilgrim_build build` (bump `--version`), publish `content.json` to the GitHub Pages manifest, and rebuild the app asset for new installs. Installed apps pull the newer manifest automatically on launch.